Introduktion til AI

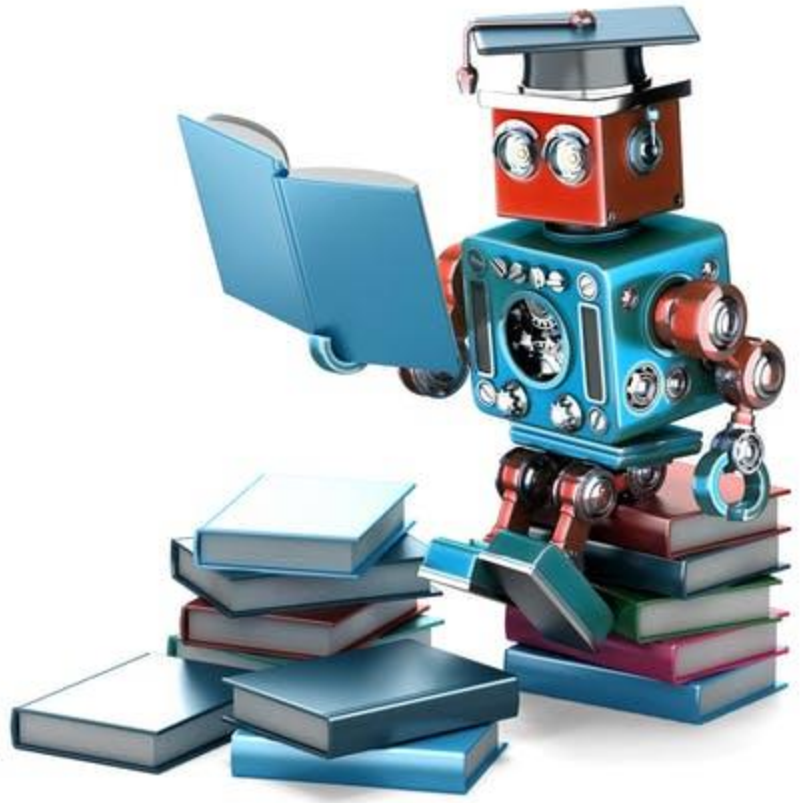


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Medical Informatics and Image Analysis

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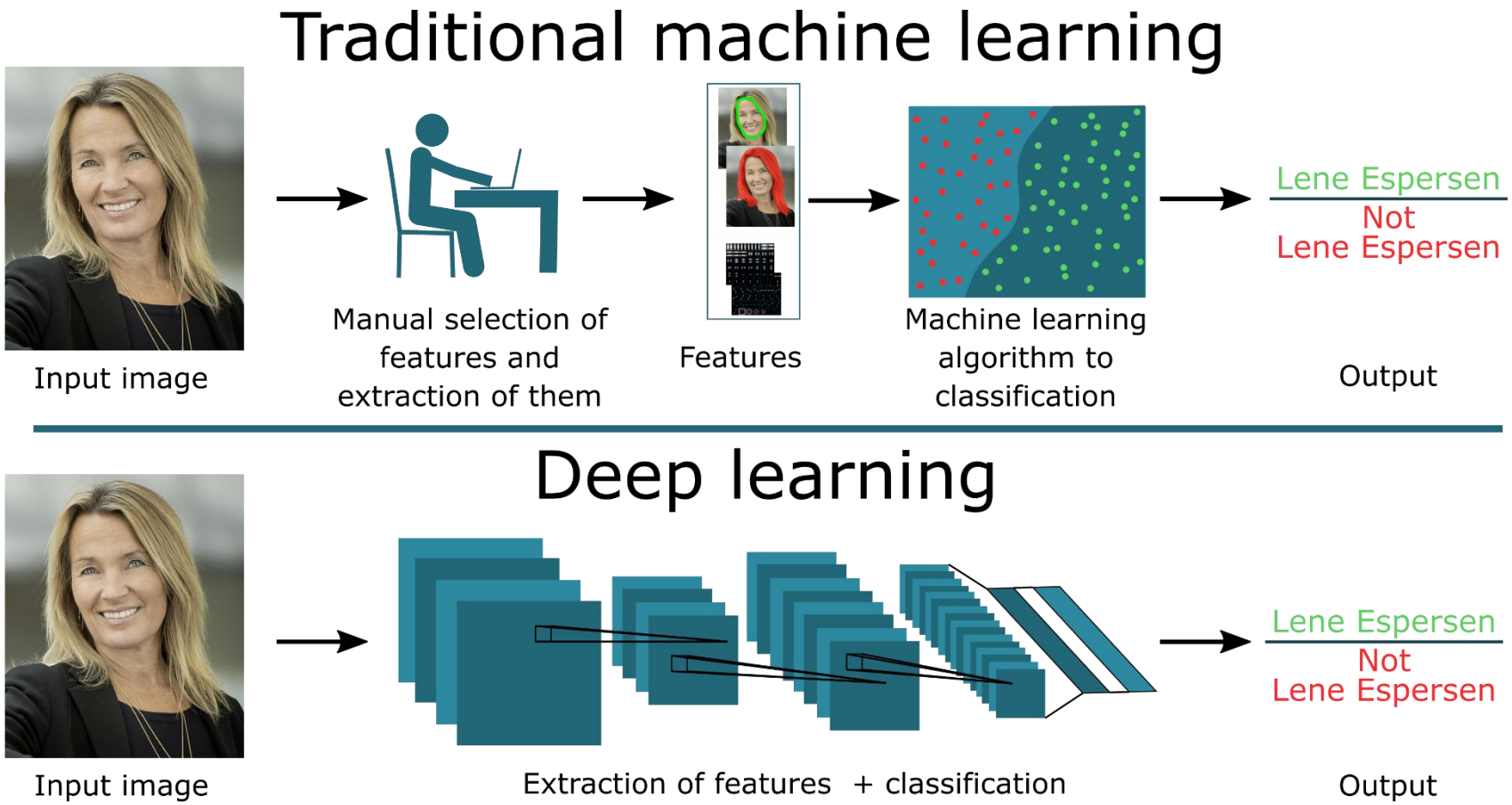
Agenda

- Intro
- Machine learning systemer
- Vigtige udfordringer ved machine learning
- Test og validering
- Opgaver



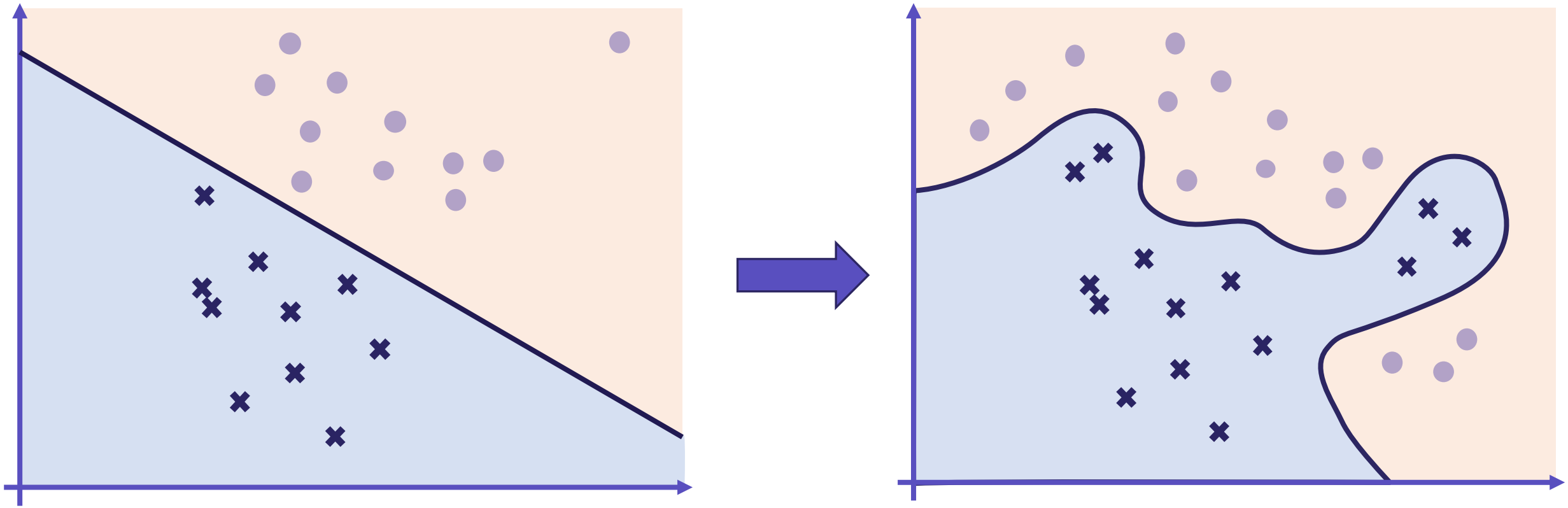
Deep learning

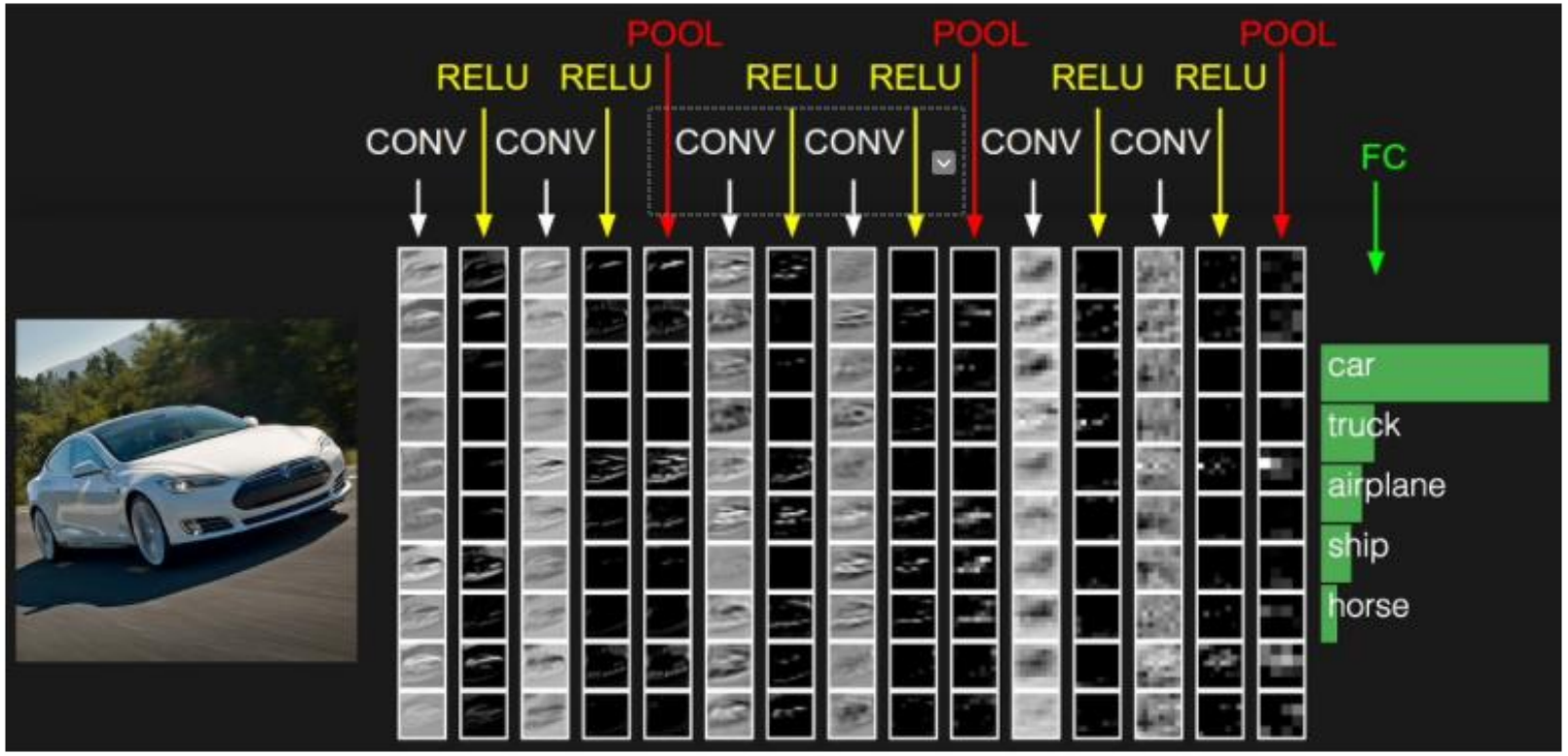
From machine learning to deep learning



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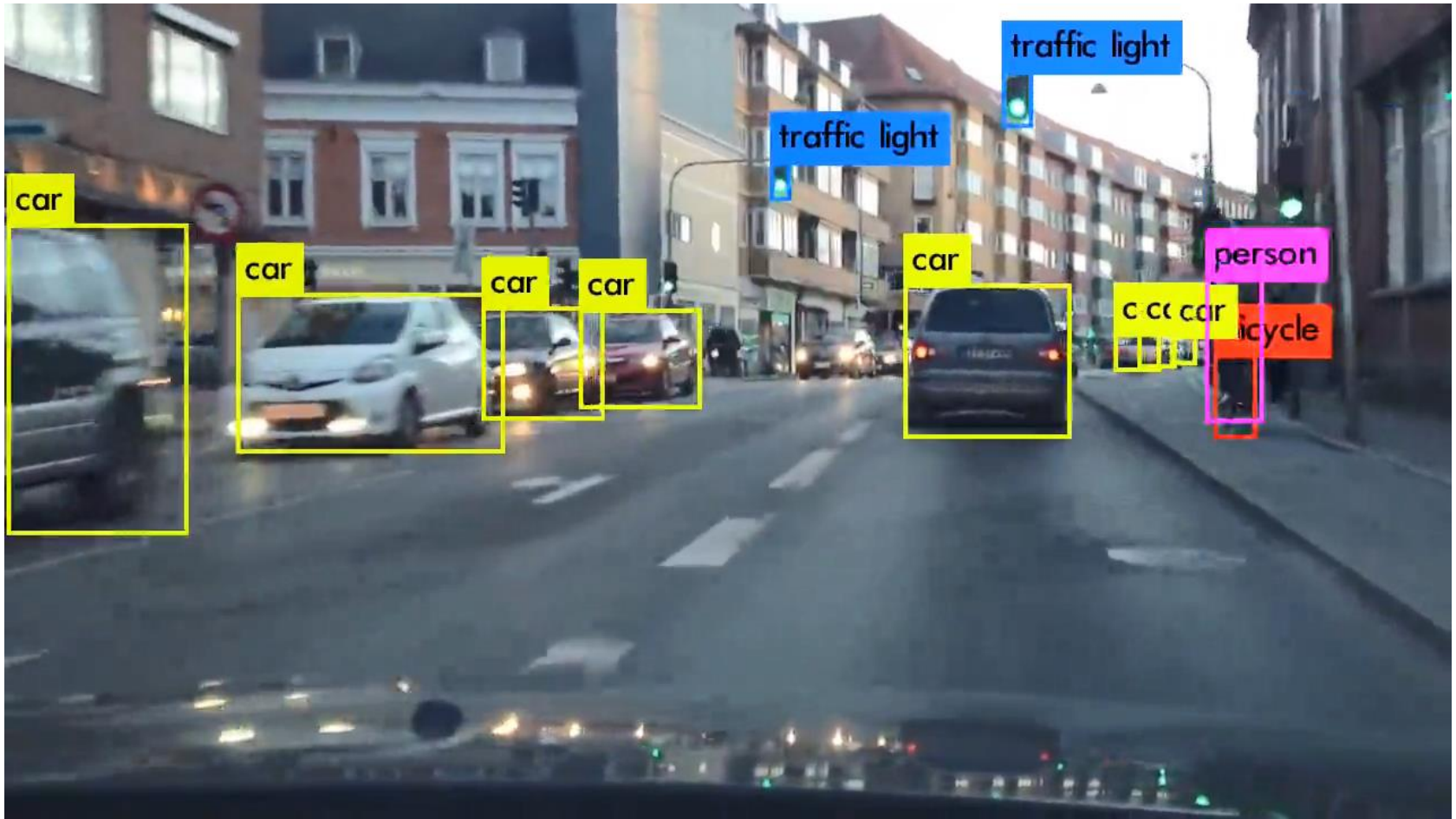
Deep learning




POOL
POOL
POOL
RELU RELU
RELU RELU
RELU RELU
CONV CONV
CONV CONV
CONV CONV
FC
car
truck
airplane
ship
horse

Deep learning

- Very impressive results!



Machine learning applications

- Optical character recognition (OCR)

- The spam filter (first mainstream ML application)
  - Examples of spam emails (flagged by user)
  - Examples of regular emails (nonspam)
  - Learns to flag spam from training data (examples)
  - A training example is called a training instance (or sample)
  - Machine learning trains a model

- Voice prompts
- Automatic translation
- Image search
- Product recommendations

Machine learning benefits

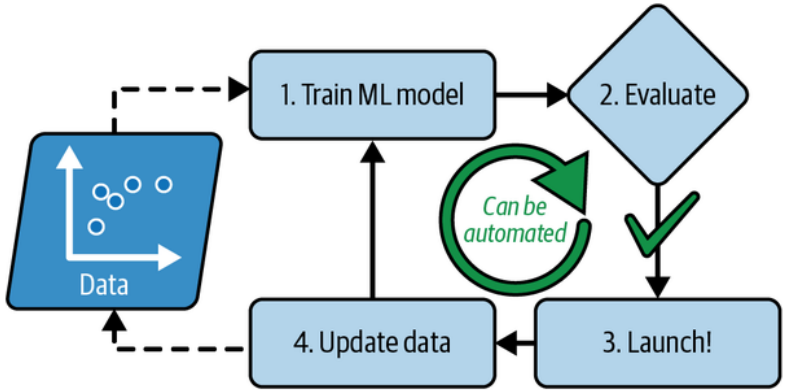


Figure 1-3. Automatically adapting to change

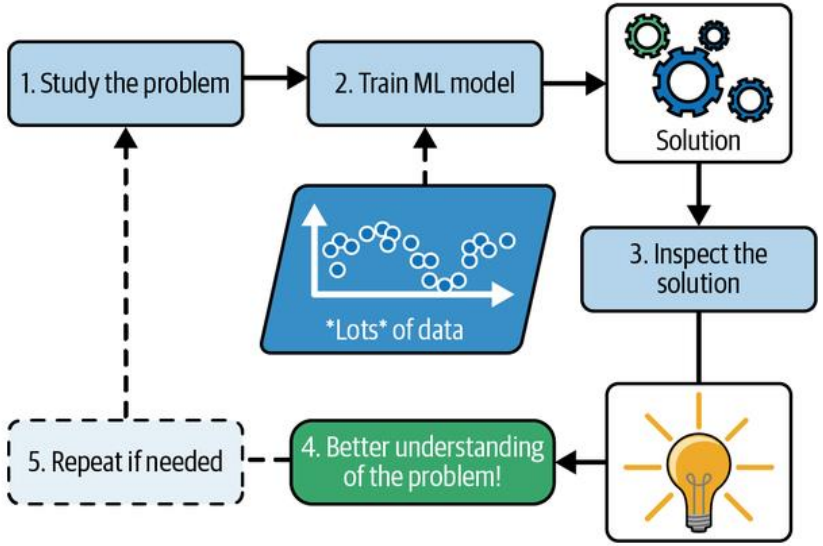


Figure 1-4. Machine learning can help humans learn

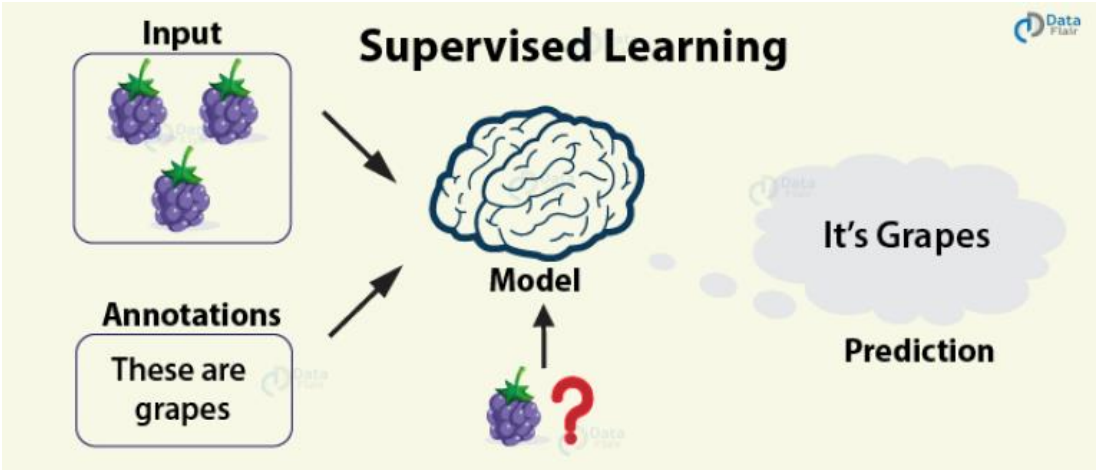
- Simplifies solutions where rules would be too long or fragile.
- Handles problems too complex for traditional approaches.
- Adapts to changing environments.
- Provides insights into complex problems and big data.

Types of Machine Learning Systems

- How they are supervised during training (supervised, unsupervised, semi-supervised, self-supervised, and others)
- Whether or not they can learn incrementally on the fly (online versus batch learning)
- Whether they work by simply comparing new data points to known data points, or instead by detecting patterns in the training data and building a predictive model, much like scientists do (instance-based versus model-based learning)

Supervised learning

- Training set includes the desired solution (labels/target value)
- Classification (spam filter) and regresssion (predict value)



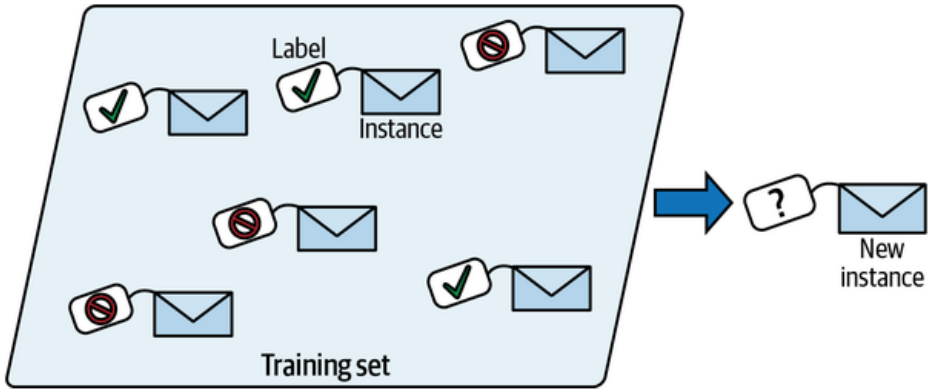


Figure 1-5. A labeled training set for spam classification (an example of supervised learning)

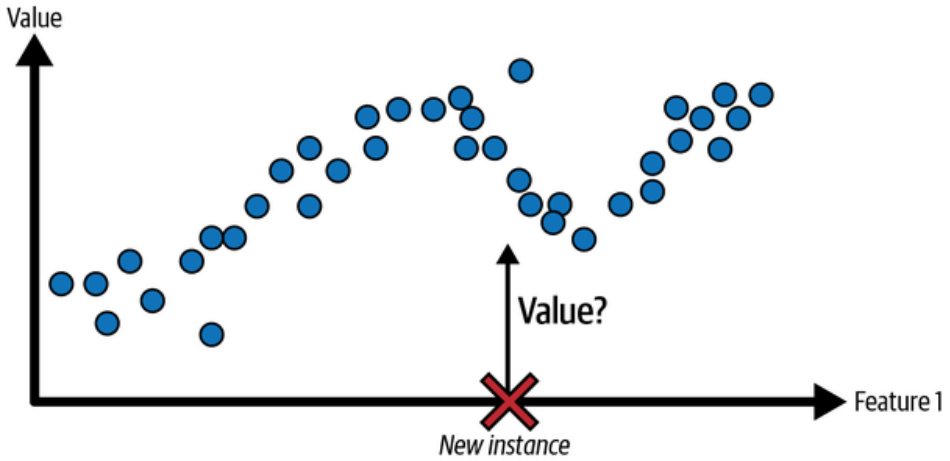
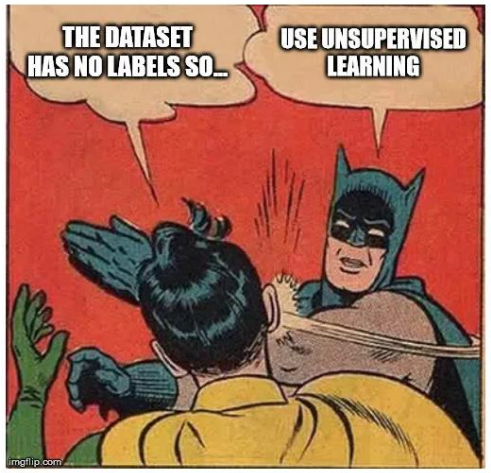
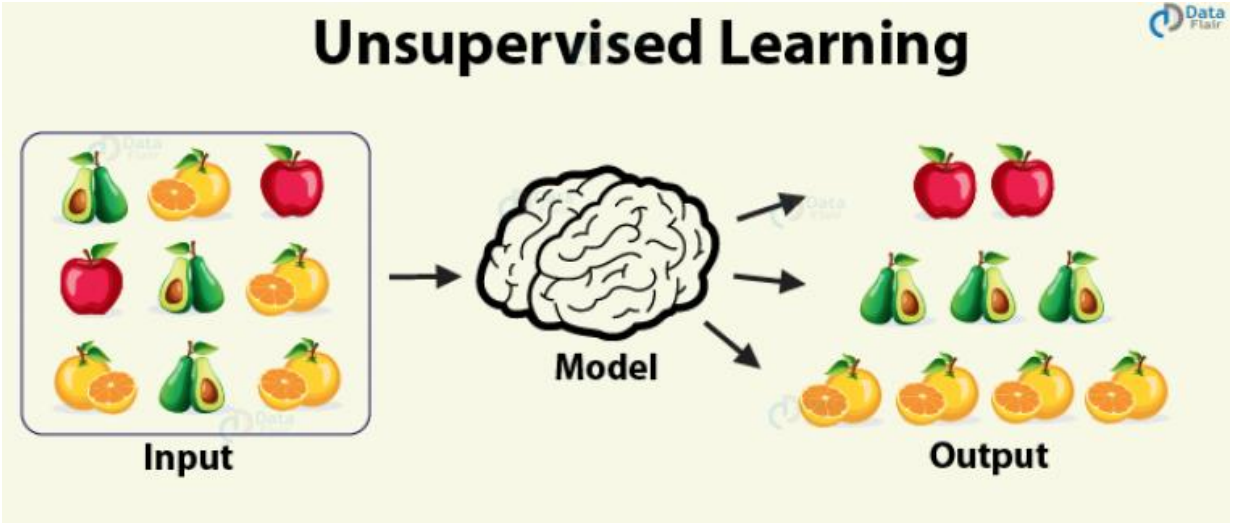


Figure 1-6. A regression problem: predict a value, given an input feature (there are usually multiple input features, and sometimes multiple output values)

Unsupervised learning

- Training data is unlabeled





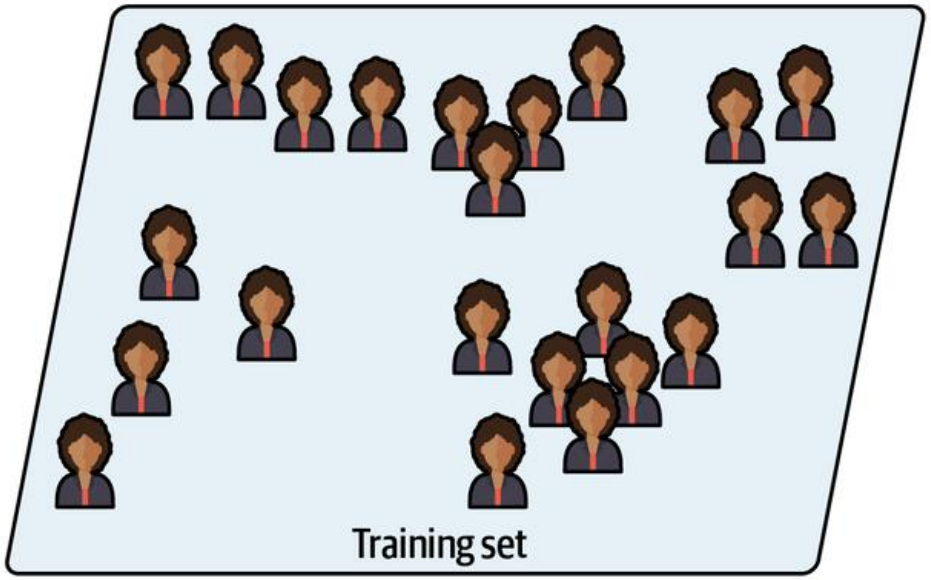


Figure 1-7. An unlabeled training set for unsupervised learning

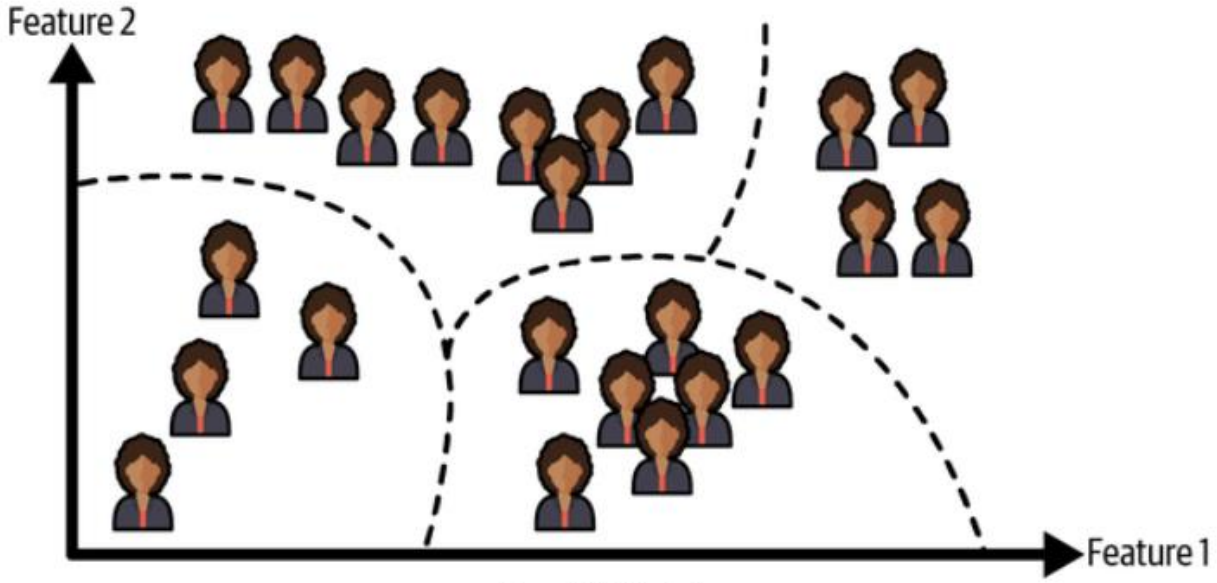


Figure 1-8. Clustering

Unsupervised learning

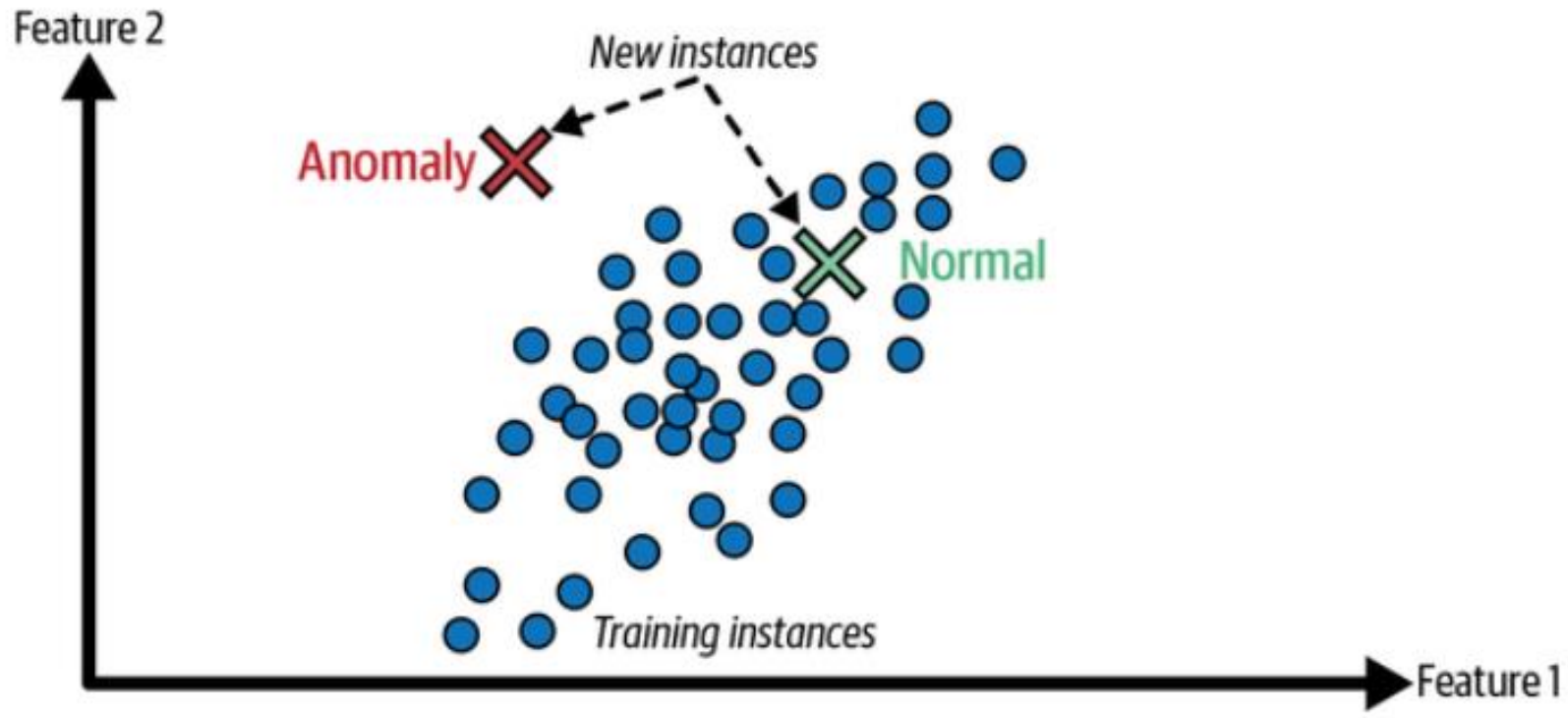


Figure 1-10. Anomaly detection

Semi-supervised learning

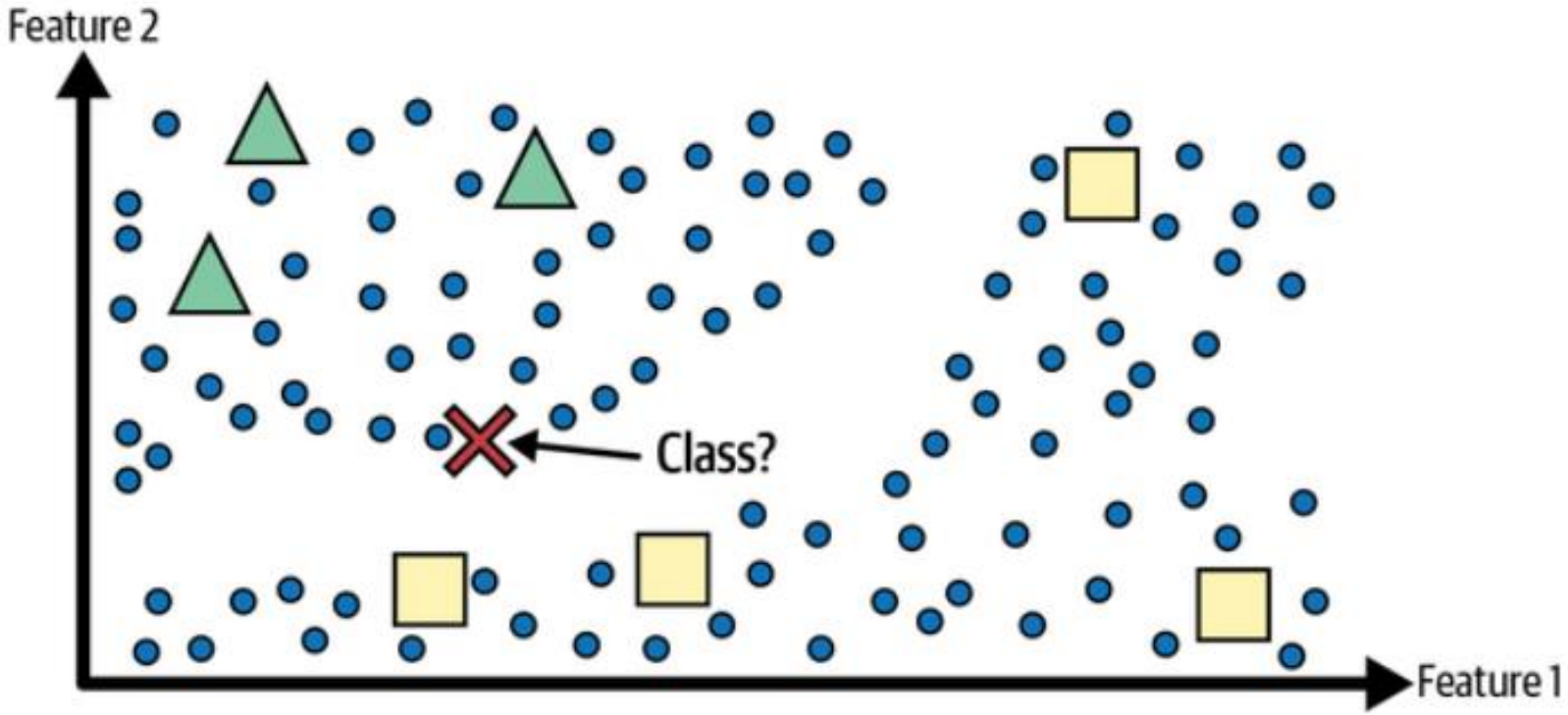
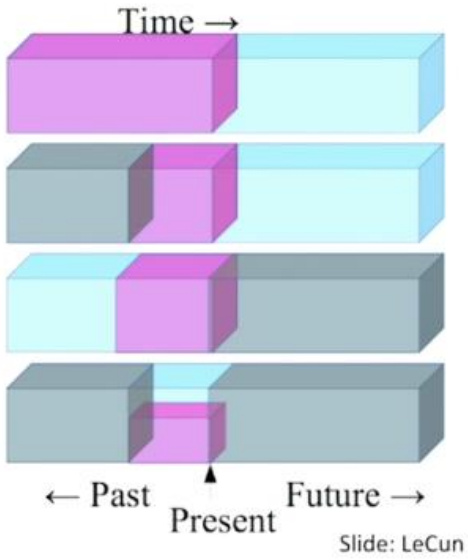


Figure 1-11. Semi-supervised learning with two classes (triangles and squares): the unlabeled examples (circles) help classify a new instance (the cross) into the triangle class rather than the square class, even though it is closer to the labeled squares

Self-supervised learning

- Predict any part of the input from any other part.
- Predict the future from the past.
- Predict the future from the recent past.
- Predict the past from the present.
- Predict the top from the bottom.
- Predict the occluded from the visible
- Pretend there is a part of the input you don't know and predict that.



Slide: LeCun

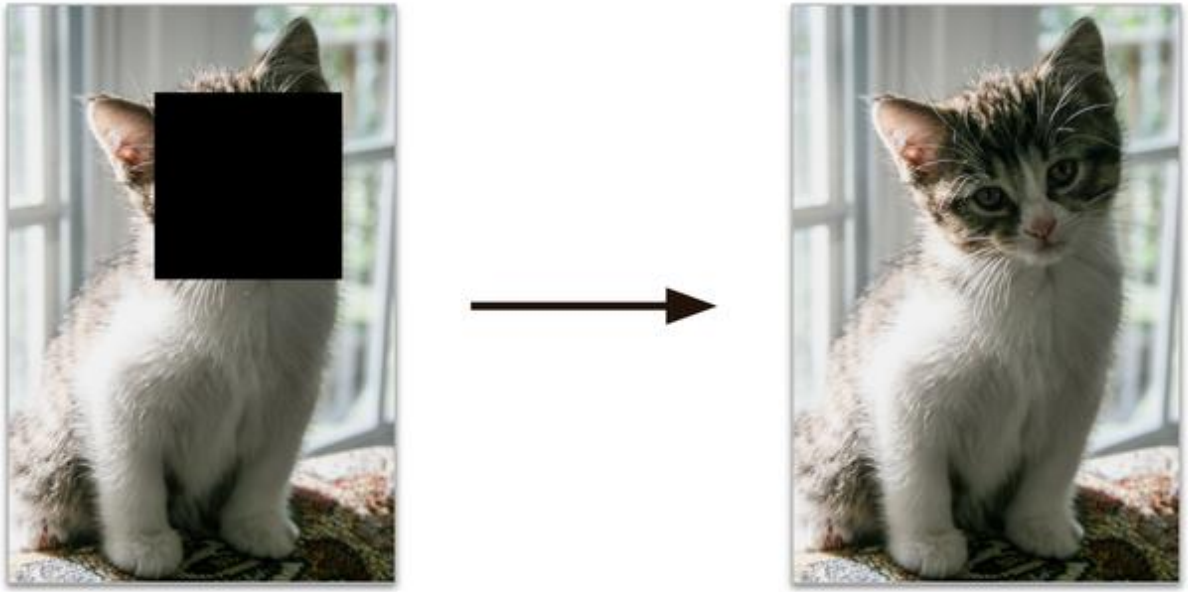
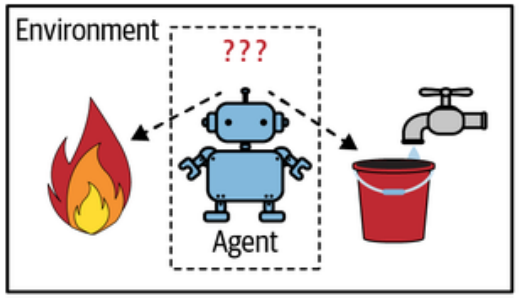
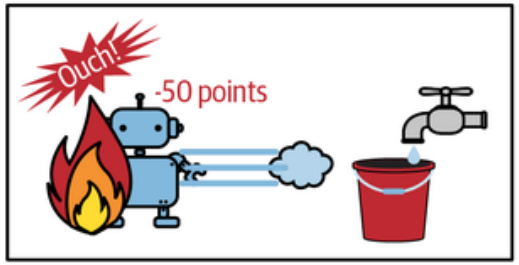


Figure 1-12. Self-supervised learning example: input (left) and target (right)

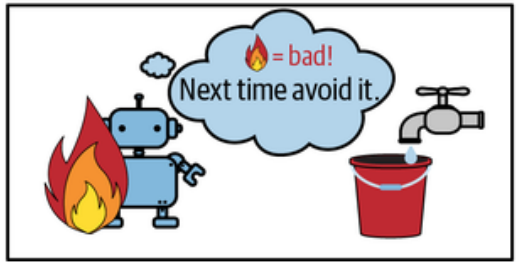
Reinforcement learning



1. Observe
2. Select action using policy



3. Action!
4. Get reward or penalty



5. Update policy (learning step)
6. Iterate until an optimal policy is found

Figure 1-13. Reinforcement learning

It works… !

- 1997: IBM's Deep Blue beats Garry Kasparov in chess



- 2016: Google's DeepMind beats Lee Sedol in GO



- 2019: Elon Musk's OpenAI beats world's top pro team in Dota 2





The Guardian

Batch learning

- Incapable of learning incrementally
- Must be trained using all the available data

1. Training offline
2. Launched into production

- Runs without learning anymore

- Model's performance tends to decay slowly over time
- The world continues to evolve while the model remains unchanged
- Model rot / data drift

- Regularly retrain the model on up-to-date data
- How often?
  - Cats and dogs (low frequency)
  - Financial market (high frequency)

Online learning

- Train the system incrementally by feeding it data instances sequentially, either individually or in small groups

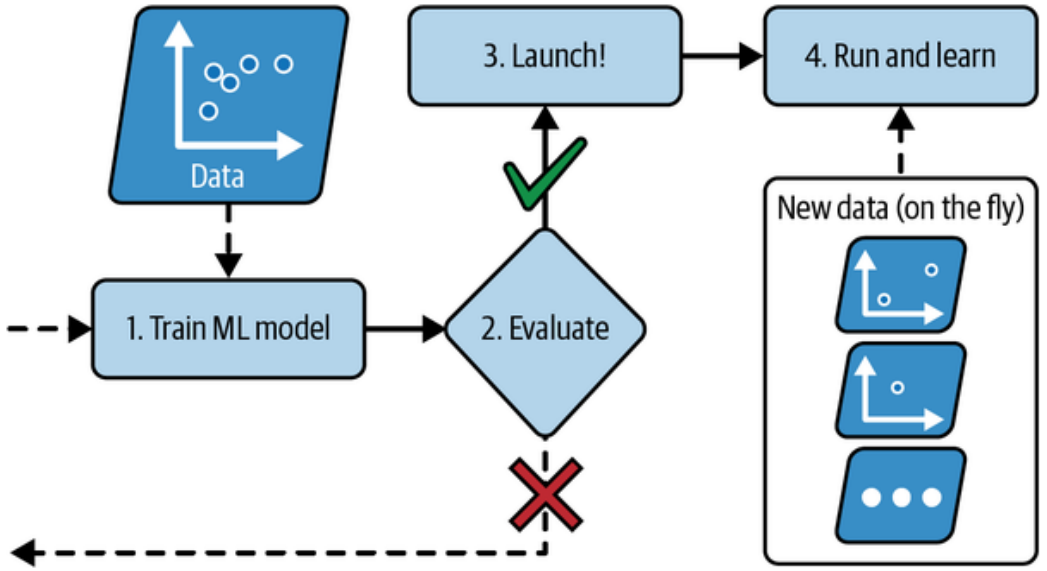


Figure 1-14. In online learning, a model is trained and launched into production, and then it keeps learning as new data comes in

Instance-Based Learning

- Classification/prediction based on identical or (very) similar instances
- Requires measure of similarity

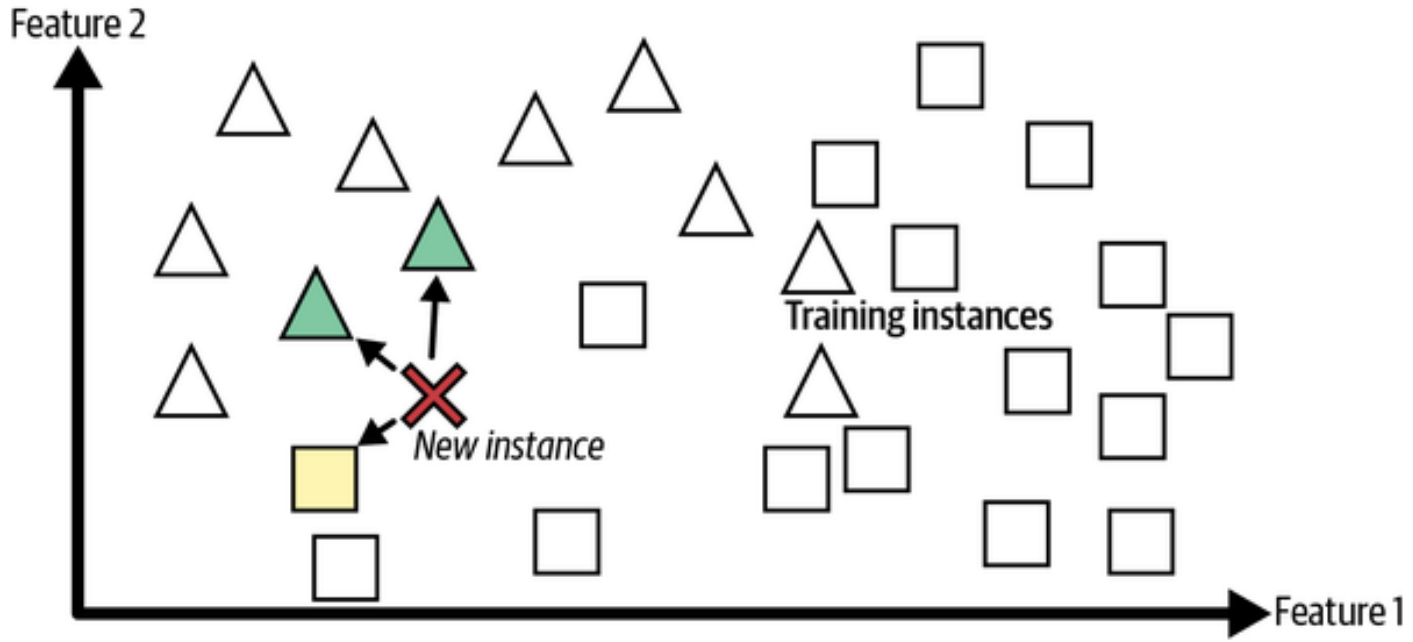


Figure 1-16. Instance-based learning

Model-based learning

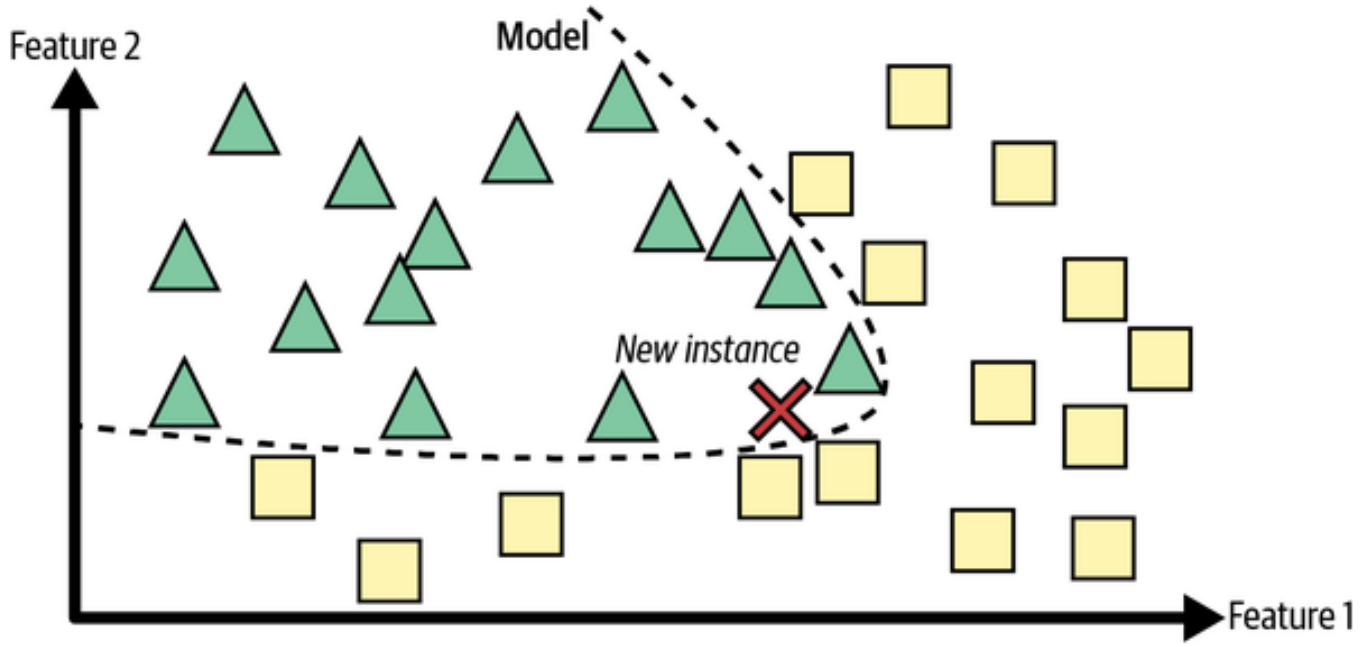


Figure 1-17. Model-based learning

Model-Based Learning

Table 1-1. Does money make people happier?

Country	GDP per capita (USD)	Life satisfaction
Turkey	28,384	5.5
Hungary	31,008	5.6
France	42,026	6.5
United States	60,236	6.9
New Zealand	42,404	7.3
Australia	48,698	7.3
Denmark	55,938	7.6

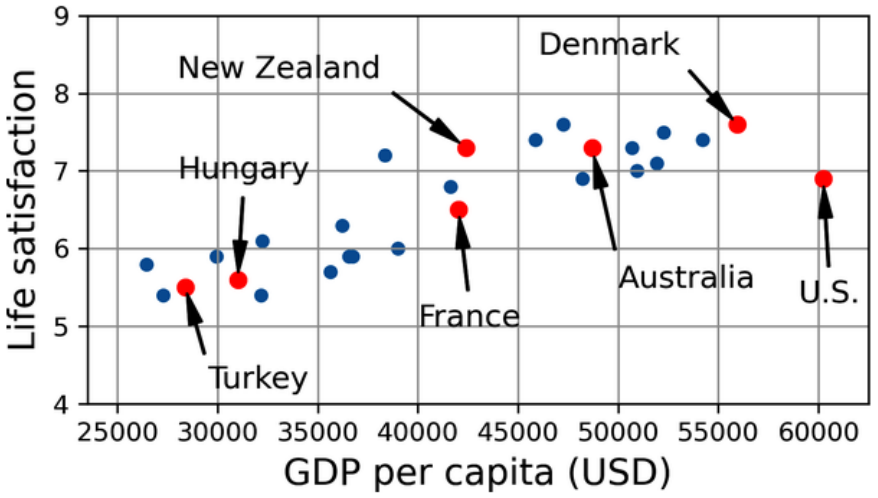


Figure 1-18. Do you see a trend here?

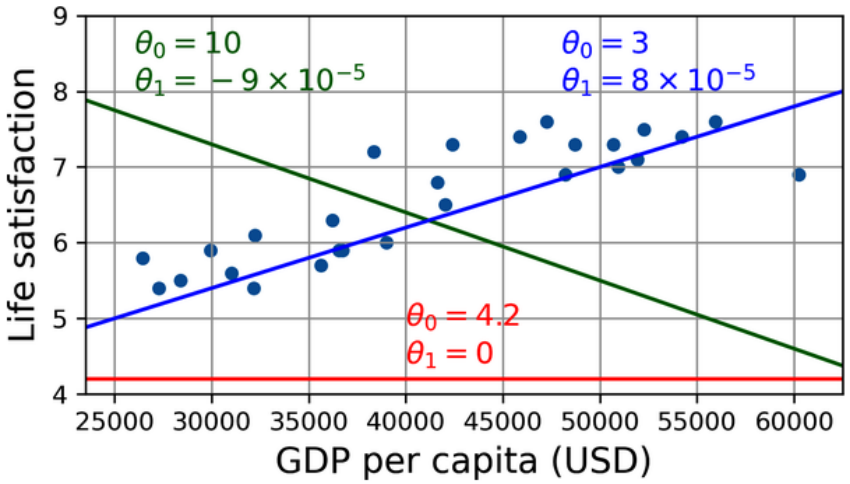


Figure 1-19. A few possible linear models

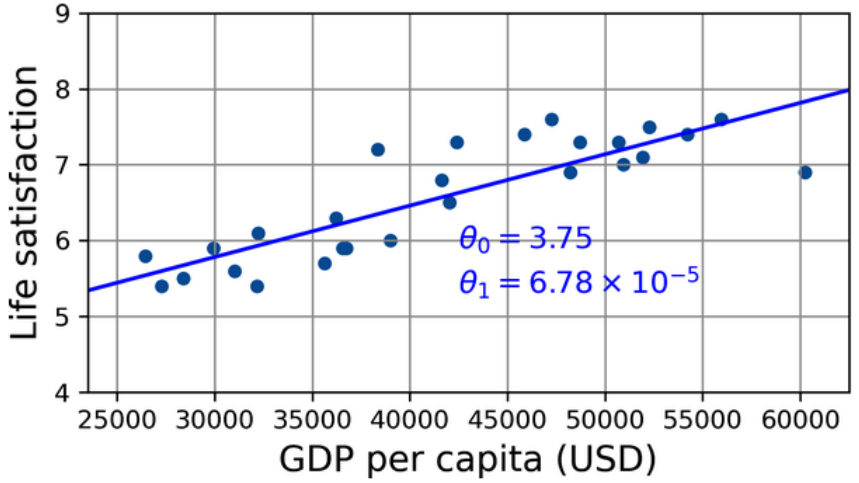


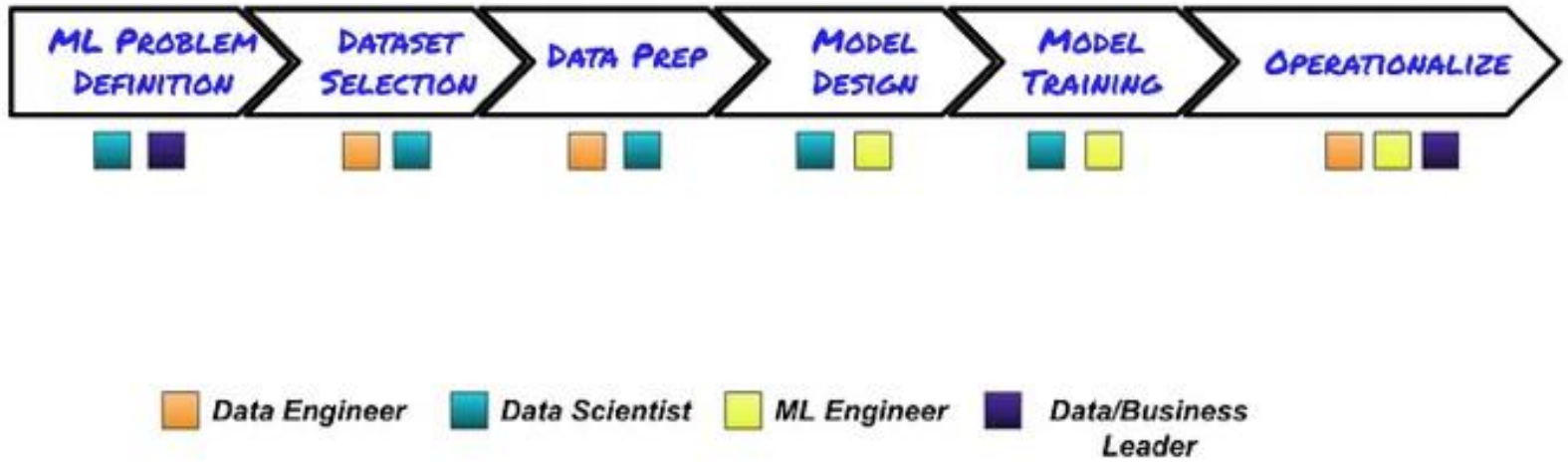
Figure 1-20. The linear model that fits the training data best

$$\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$$

What Could Possibly Go Wrong???

98 things that can go wrong in a ML project

87% of ML projects fail today!



- Bad data!!!
- Bad model!!!

Insufficient Quantity of Training Data

- Simple problems - thousands of examples
- Complex problems - millions of examples

- The Unreasonable Effectiveness of Data
  - Very different machine learning algorithms performed almost identically well on a complex problem once they were given enough data
  - Data matters more than algorithms for complex problems?
  - Small and medium-sized datasets are still very common…so don't abandon algorithms just yet!

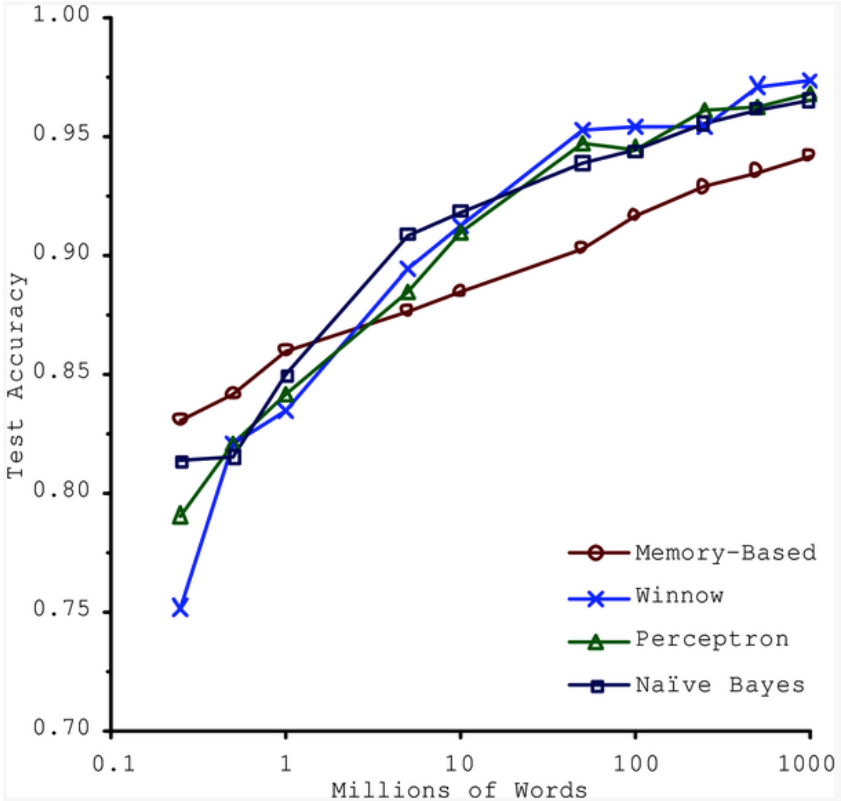


Figure 1-21. The importance of data versus algorithms[8]

Nonrepresentative Training Data

- To generalize well – data must be representative
- Inaccurate predictions
- Too small sample
- Sampling bias

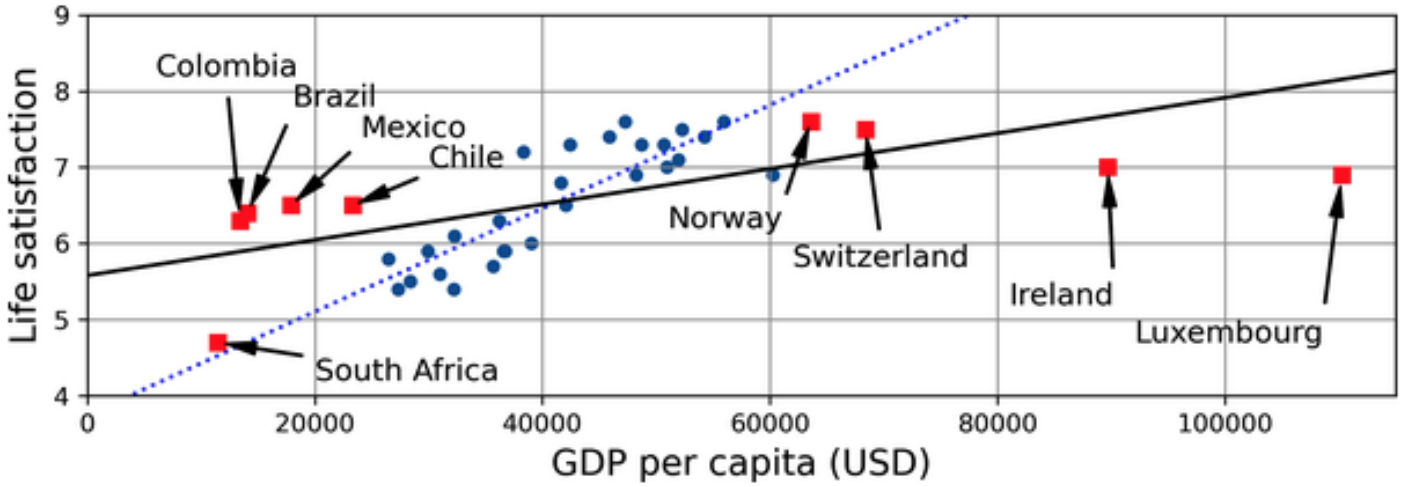


Figure 1-22. A more representative training sample

Poor-Quality Data and Irrelevant Features

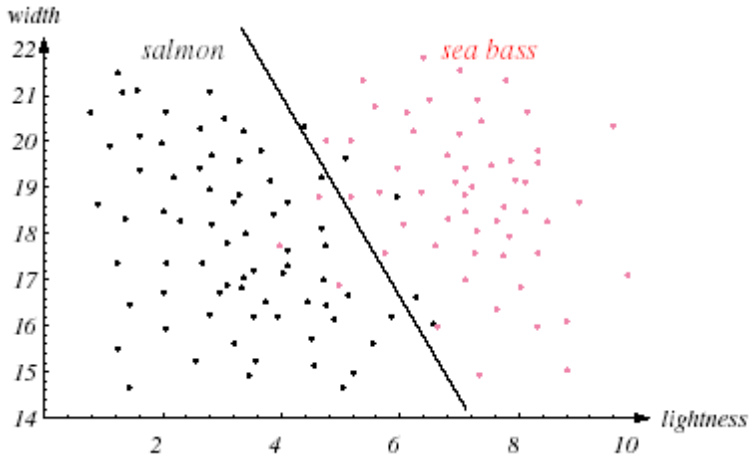
- Errors, outliers, noise – cleaning up training data
- Missing features – ignore attribute, ignore instances, fill in missing values, separate model

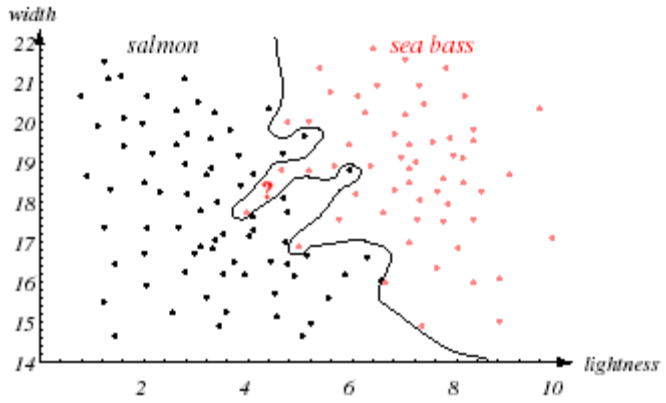
- Garbage in, garbage out
- Feature engineering – finding good features

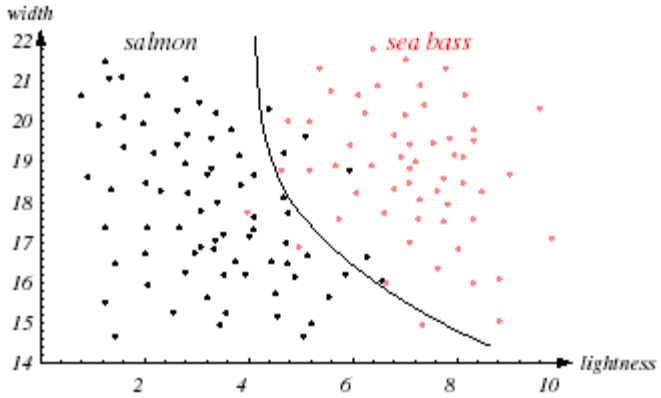
- Feature selection - select useful features
- Feature extraction – combine existing features
- Creating new features from new data



Overfitting the Training Data – Fish example







Overfitting the Training Data - Regularization

- Model performs well on the training data, but it does not generalize well
- Constrain a model to make it simpler is called regularization
- Use of hyperparameter to control the learning process
  - Number of features
  - Choice of model
  - Fixed model values (not trainable)
  - …

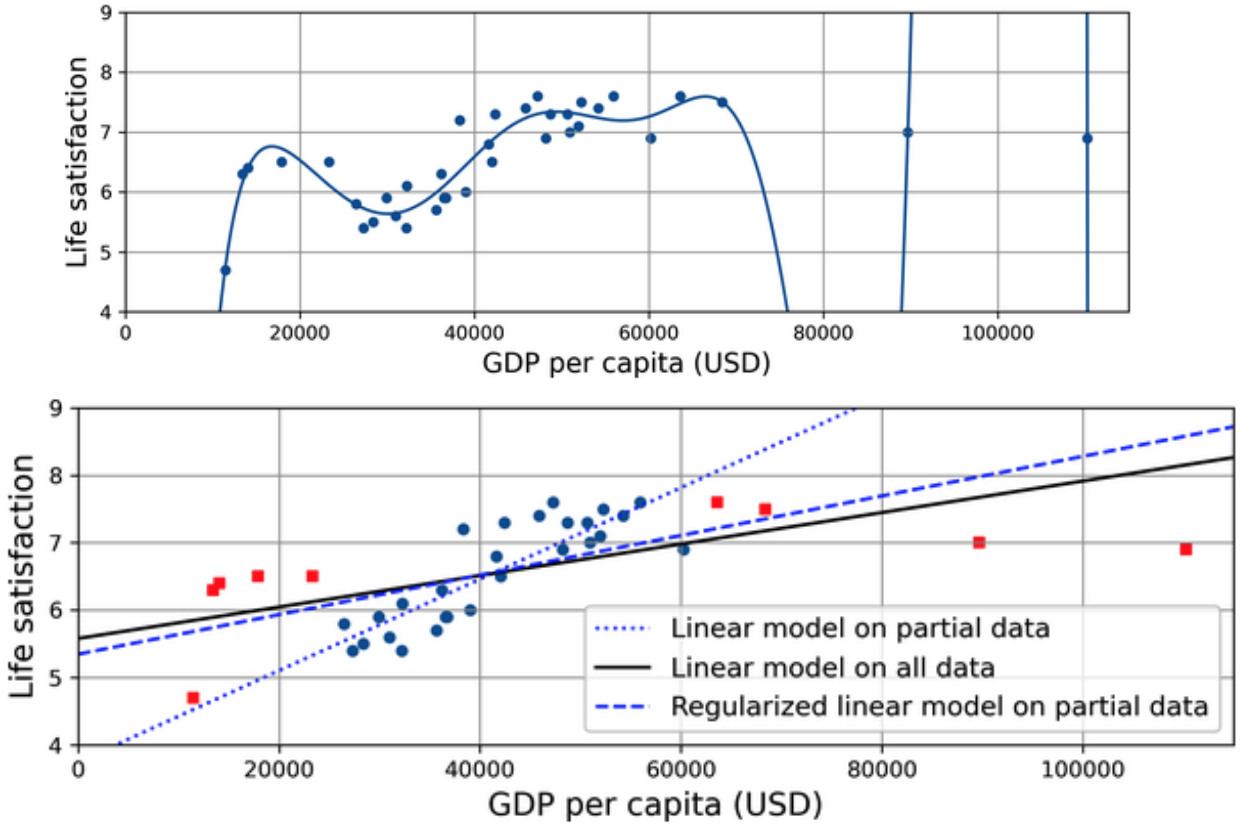


Figure 1-24. Regularization reduces the risk of overfitting

Underfitting the Training Data

- Model is too simple to learn the underlying structure of the data
- Reality is just more complex than the model!

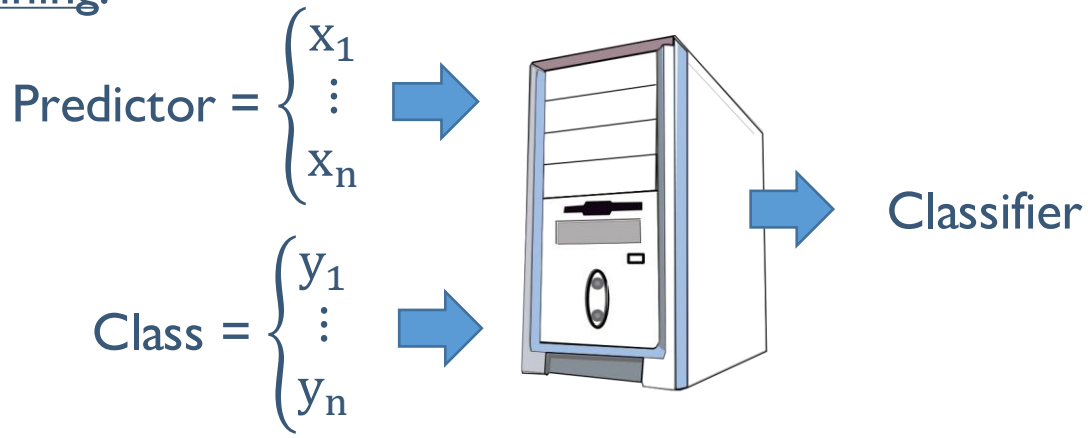
- Select a more powerful model, with more parameters.
- Feed better features to the learning algorithm (feature engineering).
- Reduce the constraints on the model (for example by reducing the regularization hyperparameter).

Testing and Validating

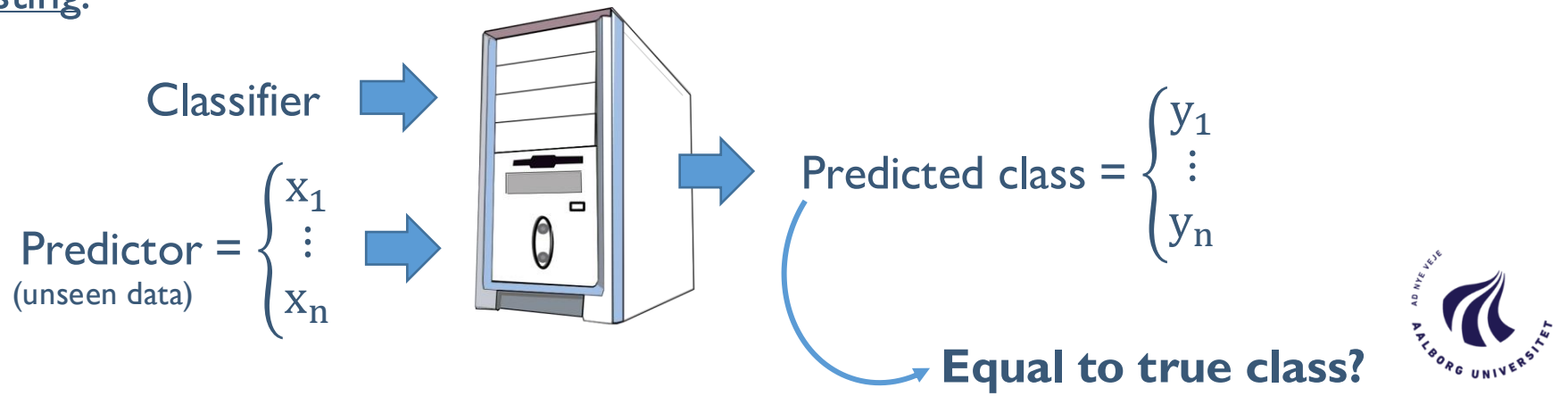
- How well will model generalize to new cases?
- Put your model in production and monitor how well it performs – NO!!!
- Split data into two sets: training set and test set (80% for training and 20% for testing is common)
- Error rate (generalization error) from test set
- Low training error + high test/generalization error = overfitting

Classifier validation

Training:



Testing:



Hyperparameter Tuning and Model Selection

- Decide between models – compare models
- Decide between model values – compare model values

- Do not base the comparison on test set ☹
- Use validation set ☺

- Holdout part of the training set for validation
- After choosing best model (and model values)
  - Train the best model on the full training set
  - Evaluate final/best model on the test set -> generalization error

- Cross-validation if training set is small

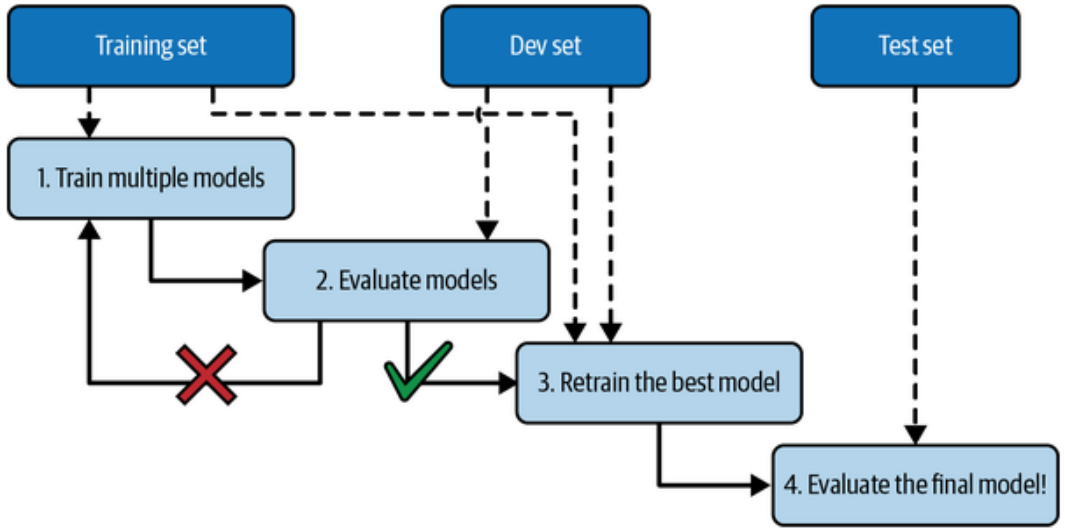
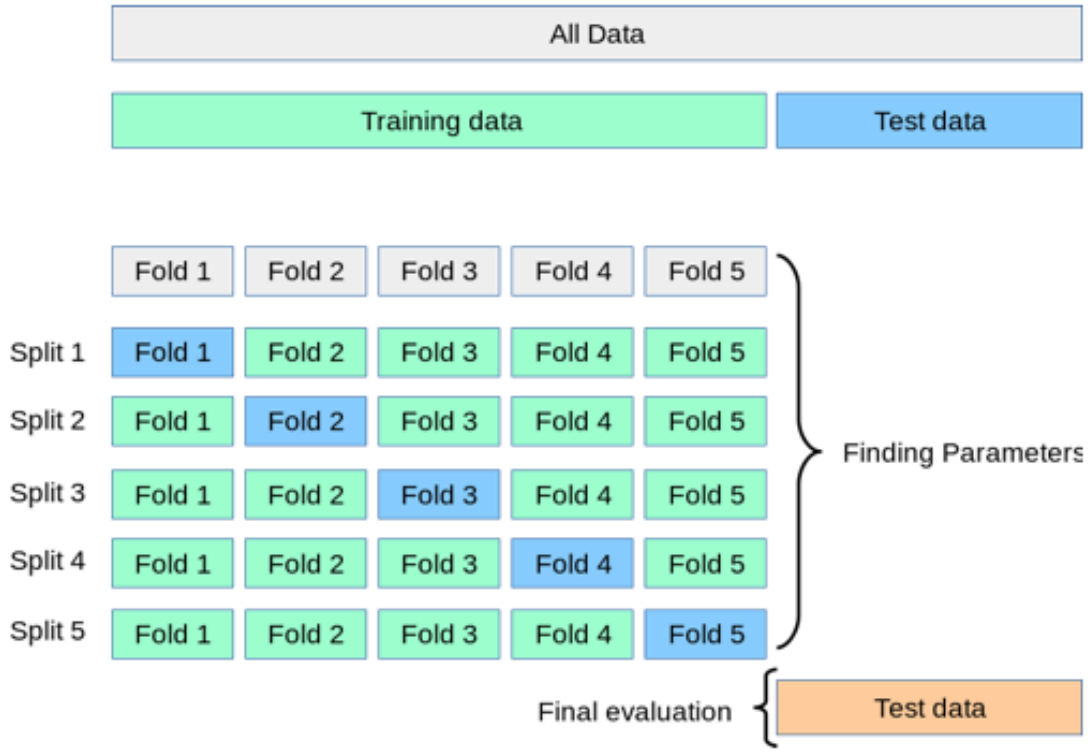


Figure 1-25. Model selection using holdout validation

Evaluate Your System on the Test Set

- Hold out test data
- Consider cross-validation if training set is small



Data Mismatch

- Training data is not representative of production data
  - Mobile app for recognition of flower species
  - Model trained on pictures from the web
  - Recognize flowers from pictures taken with mobile
- Validation set and test set must be representative (pictures taken with mobile)
- Are bad results due to overfitting or mismatch between web pictures and mobile pictures?

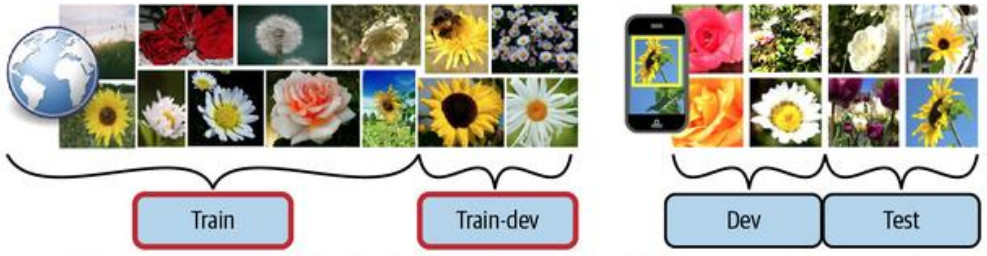


Figure 1-26. When real data is scarce (right), you may use similar abundant data (left) for training and hold out some of it in a train-dev set to evaluate overfitting; the real data is then used to evaluate data mismatch (dev set) and to evaluate the final model's performance (test set)